

Bruno L. Giordano

Auditory cognitive neuroscience

Education

- 2025 **HDR (Habilitation à diriger des recherches)**, Aix-Marseille Université, Marseille, France, thesis: “The perceptual and cerebral representation of natural sounds”.
- 2002–2005 **Ph.D., Perception and Psychophysics**, Department of General Psychology, University of Padova, Italy and Institut de Recherche et Coordination Acoustique/Musique (STMS-IRCAM-CNRS), Paris, France, thesis: “Sound Source Perception in Impact Sounds”. Advisors: Prof. Stephen McAdams and Dr. Paola Bressan.
- 1995–2001 **Degree, Experimental Psychology**, Faculty of Psychology, University of Padova, Italy, 110/110 *cum laude*, the Italian Degree (“laurea specialistica”) is equivalent to the Master’s degree. Thesis: “Psychophysics of timbre in piano tones”. Advisor: Prof. Giovanni B. Vicario.

Academic positions

- 2017– **Research Scientist (CNRS CRCN, Echelon 10)**, Institut de Neurosciences de la Timone UMR 7289, Centre National de la Recherche Scientifique, Marseille, France.
- 2015–2017 **Research Associate**, Institute of Neuroscience and Psychology (INP), University of Glasgow, UK.
- 2014–2016 **Scientific advisor**, Dr. Stefano Papetti, Institute for Computer Music and Sound Technology, Zurich University of the Arts, Switzerland.
- 2014–2015 **Postdoctoral Research Fellow**, Prof. C. Kayser, INP, University of Glasgow, UK.
- 2012–2014 **Marie Curie Research Fellow**, hosted by Prof. Pascal Belin, INP, University of Glasgow, Glasgow, UK.
- 2011–2012 **Visiting Researcher**, Prof. Pascal Belin, Centre for Cognitive Neuroimaging (CCNi), Institute of Neuroscience and Psychology, University of Glasgow, Glasgow, UK.
- 2009 **Postdoctoral Research Fellow**, Prof. Jeremy R. Cooperstock, Department of Electrical and Computer Engineering, McGill University.
- 2006–2011 **Postdoctoral Research Fellow, Associate Director of Music Perception and Cognition Laboratory (MPCL), Research Associate**, Prof. Stephen McAdams, CIRMMT, Department of Music Research, McGill University, Montréal, Canada.
- 2005 **Research Assistant**, Prof. Stephen McAdams, CIRMMT, Department of Music Research, McGill University.
- 2004 **Visiting doctoral student**, Prof. Stephen McAdams, STMS-IRCAM-CNRS, Paris, France.
- 2003–2004 **Visiting doctoral student (Marie Curie Training Site program)**, Prof. Roberto Bresin, Department of Speech, Music and Hearing (TMH), Royal Institute of Technology (KTH), Stockholm, Sweden.
- 2002 **Research Assistant**, Prof. Giovanni B. Vicario, Department of Philosophy, University of Udine, Italy.

Publications

Journal papers

- [J1] Priftis, K., & Giordano, B.L. (2026). Pure global amusia in a professional opera singer. *Neurological Sciences*, 47(5), 414. doi:10.1007/s10072-026-08976-8.
- [J2] Lamothe, C., Thoret, E., Trapeau, R., Giordano, B.L., Sein, J., Takerkart, S., Ayache, S., Artieres, T., & Belin, P. (2026). Reconstructing voice identity from noninvasive auditory cortex recordings. *eLife*, 13, RP98047. doi:10.7554/eLife.98047.
- [J3] López-Madrona, V.J., Giroud, J., Mercier, M., Lancia, L., Giordano, B.L., Trébuchon, A., Poeppel, D., Giraud, A.L., Arnal, L.H., & Morillon, B. (2026). Speech is defined by theta-gamma coupled acoustic rhythms, mapped onto segregated populations in human early auditory cortex. *eLife*, 15, RP110310. doi:10.7554/elife.110310.1.
- [J4] Esposito, M., Valente, G., Plasencia-Calaña, Y., Dumontier, M., Giordano, B.L., & Formisano, E. (2024). Bridging auditory perception and natural language processing with semantically informed deep neural networks. *Scientific Reports*, 14(1), 20994. doi:10.1038/s41598-024-71693-9.
- [J5] Cavalli, E., Chanoine, V., Tan, Y., Anton, J.L., Giordano, B.L., Pegado, F., & Ziegler, J.C. (2024). Atypical hemispheric re-organization of the reading network in high-functioning adults with dyslexia: Evidence from representational similarity analysis. *Imaging Neuroscience*. doi:10.1162/imag_a_00070.
- [J6] **Giordano, B. L.**, Esposito, M., Valente, G., & Formisano, E. (2023). Intermediate acoustic-to-semantic representations link behavioral and neural responses to natural sounds. *Nature Neuroscience*, 26, 664–672. doi:10.1038/s41593-023-01285-9. **Corresponding author.**
- [J7] Bailey, K.M., Giordano, B.L., Kaas, A.L., & Smith, F.W. (2022). Decoding sounds depicting hand-object interactions in primary somatosensory cortex. *Cerebral Cortex*. doi:10.1093/cercor/bhac296.
- [J8] **Giordano, B. L.**, de Miranda Azevedo, R., Plasencia-Calaña, Y., Formisano, E., & Dumontier, M. (2022). What do we mean with sound semantics, exactly? A survey of taxonomies and ontologies of everyday sounds. *Frontiers in Psychology*, 13. doi:10.3389/fpsyg.2022.964209. **Corresponding author.**
- [J9] Lillywhite, A., Nijhof, D., Glowinski, D., Giordano, B.L., Camurri, A., Cross, I., & Pollick, F. (2022). An fMRI examination of audiovisual observation of a point-light string quartet using intersubject correlation and physical feature analysis. *Frontiers in Neuroscience*, 16. doi:10.3389/fnins.2022.921489.
- [J10] Combrisson, E., Allegra, M., Basanisi, R., Ince, R.A., Giordano, B.L., Bastin, J., & Brovelli, A. (2022). Group-level inference of information-based measures for the analyses of cognitive brain networks from neurophysiological data. *Neuroimage*, 258, 119347. doi:10.1016/j.neuroimage.2022.119347.
- [J11] Bodin, C., Trapeau, R., Nazarian, B., Sein, J., DeGiovanni, X., Baurberg, J., Rapha, E., Renaud, L., Giordano, B.L., & Belin, P. (2021). Functionally homologous representation of vocalizations in the auditory cortex of humans and macaques. *Current Biology*, 31, 4839–4844.e4. doi:10.1016/j.cub.2021.08.043.
- [J12] **Giordano, B. L.**, Whiting, C., Kriegeskorte, N., Kotz, S.A., Gross, J., & Belin, P. (2021). The representational dynamics of perceived voice emotions evolve from categories to dimensions. *Nature Human Behaviour*, 5, 1203–1213. doi:10.1038/s41562-021-01073-0. **Corresponding author.**
- [J13] Whiting, C., Kotz, S., Gross, J., **Giordano, B. L.**, & Belin, P. (2020). The perception of caricatured emotion in voice. *Cognition*, 200, 104249. doi:10.1016/j.cognition.2020.104249. **Corresponding author and co-senior author.**
- [J14] Cao, Y., Summerfield, C., Park, H., **Giordano, B.L.**, & Kayser, C. (2019). Causal inference in the multisensory brain. *Neuron*, 102, 1076–1087. doi:10.1016/j.neuron.2019.03.043. **Co-senior author.**
- [J15] **Giordano, B. L.**, Ince, R.A.A., Gross, J., Schyns, P.G., Panzeri, S., & Kayser, C. (2017). Contributions of local speech encoding and functional connectivity to audio-visual speech perception. *eLife*, 6, e24763. doi:10.7554/eLife.24763. **Corresponding author.**
- [J16] Ince, R.A.A., Giordano, B.L., Kayser, C., Rousselet, G.A., Gross, J., & Schyns, P.G. (2017). A statistical framework for neuroimaging data analysis based on mutual information estimated via a Gaussian copula. *Human Brain Mapping*, 38, 1541–1573. doi:10.1002/hbm.23471.
- [J17] Papetti, S., Jarveläinen, H., Giordano, B.L., Schiesser, S., & Frohlich, M. (2017). Vibrotactile sensitivity in active touch: Effect of pressing force. *IEEE Transactions on Haptics*, 10, 113–122. doi:10.1109/TOH.2016.2582485.

- [J18] Cao, Y., Giordano, B.L., Avanzini, F., & McAdams, S. (2016). The dominance of haptics over audition in controlling wrist velocity during striking movements. *Experimental Brain Research*, 234, 1145–1158. doi:10.1007/s00221-015-4529-9. Co-first author and corresponding author.
- [J19] McAdams, S., Miller, M., Wild, J., & Giordano, B.L. (2015). Perception of Vicentino’s 31-tone tuning system. *Journal of Interdisciplinary Music Studies*, 8, 157–181. doi:10.4407/jims.2014.06.001.
- [J20] Gygi, B., Shafiro, V., Giordano, B.L., Kharkhurin, A., & Zhang, P.X. (2015). Predicting the timing of dynamic events through sound: Bouncing balls. *Journal of the Acoustical Society of America*, 138, 457–466. doi:10.1121/1.4923020.
- [J21] Saitis, C., Scavone, G.P., Fritz, C., & Giordano, B.L. (2015). Effect of task constraints on the perceptual evaluation of violins. *Acta Acustica united with Acustica*, 101, 382–393. doi:10.3813/AAA.918834.
- [J22] Fleming, D., Giordano, B.L., Caldara, R., & Belin, P. (2014). A language-familiarity effect for speaker discrimination without comprehension. *Proceedings of the National Academy of Sciences of the United States of America*, 111, 13795–13798. doi:10.1073/pnas.1401383111.
- [J23] Giordano, B.L., Egermann, H., & Bresin, R. (2014). The production and perception of emotionally expressive walking sounds: Similarities between musical performance and everyday motor activity. *PLoS One*, 9, e115587. doi:10.1371/journal.pone.0115587.
- [J24] **Giordano, B. L.**, Pernet, C., Charest, I., Belizaire, G., Zatorre, R.J., & Belin, P. (2014). Automatic domain-general processing of sound source identity in the left posterior middle frontal gyrus. *Cortex*, 58, 170–185. doi:10.1016/j.cortex.2014.06.005. **Corresponding author.**
- [J25] Ahrens, M.M., Hasan, B.A.S., Giordano, B.L., & Belin, P. (2014). Gender differences in the temporal voice areas. *Frontiers in Neuroscience*, 8. doi:10.3389/fnins.2014.00228.
- [J26] **Giordano, B. L.**, McAdams, S., Zatorre, R.J., Kriegeskorte, N., & Belin, P. (2013). Abstract encoding of auditory objects in cortical activity patterns. *Cerebral Cortex*, 23, 2025–2037. doi:10.1093/cercor/bhs162. **Corresponding author.**
- [J27] Fischer, M.H., Riello, M., Giordano, B.L., & Rusconi, E. (2013). Singing numbers...in cognitive space – A dual-task study of the link between pitch, space and numbers. *Topics in Cognitive Science*, 5, 354–366. doi:10.1111/tops.12017.
- [J28] **Giordano, B. L.**, Visell, Y., Yao, H.Y., Hayward, V., Cooperstock, J., & McAdams, D. (2012). Identification of walked-upon materials in auditory, kinesthetic, haptic and audio-haptic conditions. *Journal of the Acoustical Society of America*, 131, 4002–4012. doi:10.1121/1.3699205. **Corresponding author.**
- [J29] Saitis, C., Giordano, B.L., Fritz, C., & Scavone, G. (2012). Perceptual evaluation of violins: A quantitative analysis of preference judgments by experienced players. *Journal of the Acoustical Society of America*, 136, 4002–4012. doi:10.1121/1.4765081.
- [J30] **Giordano, B. L.**, Guastavino, C., Murphy, E., Ogg, M., Smith, B.K., & McAdams, S. (2011). Comparison of methods for collecting and modeling dissimilarity data: applications to complex sound stimuli. *Multivariate Behavioral Research*, 46, 779–811. **Corresponding author.**
- [J31] Gingras, B.T., Lagrandeur-Ponce, T., Giordano, B.L., & McAdams, S. (2011). Perceiving musical identity: Performer identification is dependent on performer expertise and expressiveness, but not on listener expertise. *Perception*, 40, 1206–1220. doi:10.1068/p6891.
- [J32] Peeters, G., **Giordano, B.L.**, Susini, P., Misdariis, N., & McAdams, S. (2011). The timbre toolbox: Extracting acoustic descriptors from musical signals. *Journal of the Acoustical Society of America*, 130, 2902–2916. doi:10.1121/1.3642604. **Co-first author.**
- [J33] Visell, Y., Giordano, B.L., Millet, G., & Cooperstock, J. (2011). Vibration influences haptic perception of surface compliance during walking. *PLoS One*, 6, e17697. doi:10.1371/journal.pone.0017697.
- [J34] **Giordano, B. L.**, McDonnell, J., & McAdams, S. (2010). Hearing living symbols and nonliving icons: Category-specificities in the cognitive processing of environmental sounds. *Brain and Cognition*, 73, 7–19. doi:10.1016/j.bandc.2010.01.005. **Corresponding author.**
- [J35] **Giordano, B. L.**, Rocchesso, D., & McAdams, S. (2010). Integration of acoustical information in the perception of impacted sound sources: The role of information accuracy and exploitability. *Journal of Experimental Psychology: Human Perception and Performance*, 36, 462–479. doi:10.1037/a0018388. **Corresponding author.**
- [J36] **Giordano, B. L.**, & McAdams, S. (2010). Sound source mechanics and musical timbre perception: Evidence from previous studies. *Music Perception*, 28, 155–168. doi:10.1525/mp.2010.28.2.155. **Corresponding author.**

- [J37] Castiello, U., Giordano, B.L., Begliomini, C., Ansuini, C., & Grassi, M. (2010). When ears drive hands: The influence of contact sound on reaching to grasp. *PLoS One*, 5, e12240. doi:10.1371/journal.pone.0012240.
- [J38] McAdams, S., Roussarie, V., Chaigne, A., & Giordano, B.L. (2010). The psychomechanics of simulated sound sources: Material properties of impacted thin plates. *Journal of the Acoustical Society of America*, 128, 1401–1413. doi:10.1121/1.3466867.
- [J39] McAdams, S., & Giordano, B.L. (2010). La psychoacoustique dévoile le potentiel musical du timbre. *Acoustique & Techniques*, 60, 13–20.
- [J40] Visell, Y., Fontana, F., Giordano, B.L., Nordahl, R., Serafin, S., & Bresin, R. (2009). Sound design and perception in walking interactions. *International Journal of Human-Computer Studies*, 67, 947–959. doi:10.1016/j.ijhcs.2009.07.007.
- [J41] Rusconi, E., Kwan, B., Giordano, B.L., Umiltà, C., & Butterworth, B. (2006). Spatial representation of pitch height: the SMARC effect. *Cognition*, 99, 113–129. doi:10.1016/j.cognition.2005.01.004.
- [J42] **Giordano, B. L.**, & McAdams, S. (2006). Material identification of real impact sounds: effects of size variation in steel, glass, wood and plexiglass plates. *Journal of the Acoustical Society of America*, 119, 1171–1181. doi:10.1121/1.2149839. **Corresponding author.**
- [J43] Rusconi, E., Kwan, B., Giordano, B.L., Umiltà, C., & Butterworth, B. (2005). The Mental Space of Pitch Height. *Annals of The New York Academy Of Sciences*, 1060, 195–197. doi:10.1196/annals.1360.056.

Book chapters

- [B1] Fontana, F., Papetti, S., Järveläinen, H., Avanzini, F., & Giordano, B.L. (2018). Perception of vibrotactile cues in musical performance. In S. Papetti, & C. Saitis (Eds.), *Musical Haptics*, 49–72. Springer International Publishing. doi:10.1007/978-3-319-58316-7_4.
- [B2] Giordano, B.L., & Avanzini, F. (2014). Perception and synthesis of sound-generating materials. In M. Di Luca (Ed.), *Multisensory softness – Perceived compliance from multiple sources of information*, 49–84. London, UK: Springer. doi:10.1007/978-1-4471-6533-0_4.
- [B3] Giordano, B.L., Susini, P., & Bresin, R. (2013). Perceptual evaluation of sound-producing objects. In K. Franinovic, & S. Serafin (Eds.), *Sonic Interaction Design*, 151–198. Boston, MA: MIT Press. doi:10.7551/mitpress/8555.003.0007.
- [B4] McAdams, S., & Giordano, B.L. (2009). The perception of musical timbre. In S. Hallam, I. Cross, & M. Thaut (Eds.), *The Oxford handbook of music psychology*, 72–80. New York, NY: Oxford University Press. doi:10.1093/oxfordhb/9780198722946.013.12.
- [B5] Giordano, B.L. (2003). Everyday listening: An annotated bibliography. In D. Rocchesso, & F. Fontana (Eds.), *The sounding object*, 1–13. Firenze: Mondo Estremo.
- [B6] Giordano, B.L. (2003). Material categorization and hardness scaling in real and synthetic impact sounds. In D. Rocchesso, & F. Fontana (Eds.), *The sounding object*, 73–93. Firenze: Mondo Estremo.

Book reviews

- [BR1] Giordano, B.L. (2011). Music Perception (Springer Handbook of Auditory Research). *Journal of the Acoustical Society of America*, 129, 4086. doi:10.1121/1.3583539.

Preprints

- [PR1] Plegat, M., Araújo Vitória, M., Marinato, G., Tita, B., van der Lans, C., Pijfers, M., Esposito, M., Bertovic, M.S., Formisano, E., & Giordano, B.L. (2026). A behaviourally normed database of 1,377 natural sounds for auditory cognition and neuroscience. *bioRxiv*. doi:10.64898/2026.08.25.746933.
- [PR2] Esposito, M., Weidler, T., Ferreyra, C., Giordano, B.L., & Formisano, E. (2026). Multiscale temporal processing supports sound recognition under causal constraints. *bioRxiv*. doi:10.64898/2026.07.27.740874.
- [PR3] López-Madrona, V.J., Giroud, J., Mercier, M., Lancia, L., Giordano, B.L., Trébuchon, A., Poeppel, D., Giraud, A.L., Arnal, L.H., & Morillon, B. (2025). Speech is defined by theta-gamma coupled acoustic rhythms, mapped onto segregated populations in human early auditory cortex. *bioRxiv*. doi:10.1101/2025.10.22.683926.
- [PR4] Priftis, K., & Giordano, B.L. (2025). Pure global amusia in a professional opera singer. *OSF Preprints*. doi:10.31219/osf.io/vxm7_v1.
- [PR5] Esposito, M., Valente, G., Plasencia-Calaña, Y., Dumontier, M., Giordano, B.L., & Formisano, E. (2024). Bridging auditory perception and natural language processing with semantically informed deep neural networks. *bioRxiv*. doi:10.1101/2024.04.29.591634.

- [PR6] Lamothe, C., Thoret, E., Trapeau, R., Giordano, B., Sein, J., Takerkart, S., Ayache, S., Artieres, T., & Belin, P. (2024). Reconstructing voice identity from noninvasive auditory cortex recordings. *BioRxiv*, 2024–02.
- [PR7] Combrisson, E., Allegra, M., Basanisi, R., Ince, R.A., Giordano, B.L., Bastin, J., & Brovelli, A. (2021). Group-level inference of information-based measures for the analyses of cognitive brain networks from neurophysiological data. *bioRxiv*. doi:10.1101/2021.08.14.456339.
- [PR8] Bailey, K.M., Giordano, B.L., Kaas, A.L., & Smith, F.W. (2019). Decoding the sound of hand-object interactions in primary somatosensory cortex. *bioRxiv*. doi:10.1101/732669.
- [PR9] Cao, Y., Summerfield, C., Park, H., Giordano, B., & Kayser, C. (2018). Causal inference in the multisensory brain. *bioRxiv*, 500413. doi:10.1101/500413.
- [PR10] Giordano, B.L., Whiting, C., Kriegeskorte, N., Kotz, S.A., Belin, P., & Gross, J. (2018). From categories to dimensions: spatio-temporal dynamics of the cerebral representations of emotion in voice. *bioRxiv*, 265843. doi:10.1101/265843.
- [PR11] Giordano, B.L., Ince, R.A.A., Gross, J., Panzeri, S., Schyns, P.G., & Kayser, C. (2016). Contributions of local speech encoding and functional connectivity to audio-visual speech integration. *bioRxiv*. doi:10.1101/097493.
- [PR12] Ince, R.A.A., Giordano, B.L., Kayser, C., Rousselet, G.A., Gross, J., & Schyns, P.G. (2016). A statistical framework for neuroimaging data analysis based on mutual information estimated via a Gaussian copula. *bioRxiv*. doi:10.1101/043745.

Datasets

- [D1] Plegat, M., Araújo Vitória, M., Marinato, G., Tita, B., van der Lans, C., Pijfers, M., Esposito, M., Bertovic, M.S., Formisano, E., & Giordano, B.L. (2026). MaMa Sounds: A behaviourally normed database of 1,377 natural sounds for auditory cognition and neuroscience. doi:10.5281/zenodo.22042881.
- [D2] Priftis, K., & Giordano, B.L. (2025). Pure global amusia. doi:10.31219/osf.io/mz5nh.
- [D3] Giordano, B.L., Esposito, M., Valente, G., & Formisano, E. (2023). Data from: Intermediate acoustic-to-semantic representations link behavioural and neural responses to natural sounds. doi:10.5061/dryad.0p2ngf258.
- [D4] Giordano, B.L., de Miranda Azevedo, R., Plasencia-Calaña, Y., Formisano, E., & Dumontier, M. (2022). Data from: What do we mean with sound semantics, exactly? A survey of taxonomies and ontologies of everyday sounds. doi:10.6084/m9.figshare.20813626.v1.
- [D5] Bodin, C., Trapeau, R., Nazarian, B., Sein, J., DeGiovanni, X., Baurberg, J., Rapha, E., Renaud, L., Giordano, B.L., & Belin, P. (2021). Macaque data - functionally homologous representation of vocalizations in the auditory cortex of humans and macaques. *Zenodo repository*. doi:10.5281/zenodo.5074859.
- [D6] Giordano, B.L., Whiting, C., Kriegeskorte, N., Kotz, S.A., Gross, J., & Belin, P. (2021). Data from: The representational dynamics of perceived voice emotions evolve from categories to dimensions. doi:10.5061/dryad.m905qfv0k.
- [D7] Whiting, C., Kotz, S., Gross, J., Giordano, B., & Belin, P. (2020). Data from: The perception of caricatured emotion in voice. doi:10.5061/dryad.18931zcsn.
- [D8] Giordano, B.L., Ince, R.A.A., Gross, J., Schyns, P.G., Panzeri, S., & Kayser, C. (2017). Data from: Contributions of local speech encoding and functional connectivity to audio-visual speech perception. doi:10.5061/dryad.j4567.

Articles in conference proceedings

- [P1] Esposito, M., Weidler, T., Ferreyra, C., Giordano, B.L., & Formisano, E. (2026). Causal multiscale models of temporally resolved auditory processing in human cortex. In *9th Annual Conference on Cognitive Computational Neuroscience (CCN 2026)*. New York, NY, USA. Extended abstract.
- [P2] Ferreyra, C., Plegat, M., Marinato, G., Araújo Vitória, M., Esposito, M., Formisano, E., Artieres, T., & Giordano, B.L. (2025). Disentangling redundant and synergistic interactions in the alignment between auditory brains and machines. In *2025 Conference on Cognitive Computational Neuroscience*. Amsterdam, the Netherlands.
- [P3] Plegat, M., Marinato, G., Ferreyra, C., Araújo Vitória, M., Esposito, M., Schön, D., Formisano, E., & Giordano, B.L. (2025). Temporal dynamics of natural sounds representation in the human brain. In *2025 Conference on Cognitive Computational Neuroscience*. Amsterdam, the Netherlands.
- [P4] Esposito, M., Giordano, B.L., Valente, G., & Formisano, E. (2024). Evaluating the impact of multiscale temporal processing on sound-to-event recurrent neural networks. In *2024 Conference on Cognitive Computational Neuroscience*. Boston, Massachusetts, USA.

- [P5] Araújo Vitória, M., Plegat, M., Marinato, G., Esposito, M., Herff, C., Giordano, B.L., & Formisano, E. (2024). Optimal stimulus selection for dissociating acoustic and semantic processing of natural sounds. In *2024 Conference on Cognitive Computational Neuroscience*. Boston, Massachusetts, USA.
- [P6] Wijngaard, G., Formisano, E., Giordano, B.L., & Dumontier, M. (2023). ACES: evaluating automated audio captioning models on the semantics of sounds. In *31st European Signal Processing Conference (EUSIPCO-2023)*. Helsinki, Finland.
- [P7] Esposito, M., Valente, G., Plasencia-Calaña, Y., Dumontier, M., Giordano, B.L., & Formisano, E. (2023). Semantically-informed deep neural networks for sound recognition. In *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 1–5. doi:10.1109/ICASSP49357.2023.10095606.
- [P8] Daube, C., Giordano, B.L., Schyns, P., & Ince, R.A.A. (2019). Quantitatively comparing predictive models with the partial information decomposition. In *2019 Conference on Cognitive Computational Neuroscience*. Berlin, Germany. doi:10.32470/CCN.2019.1142-0.
- [P9] Cao, Y., Park, H., Giordano, B.L., Kayser, C., Spence, C., & Summerfield, C. (2019). Unfolding of multi-sensory inference in the brain and behavior. In *2019 Conference on Cognitive Computational Neuroscience*. Berlin, Germany. doi:10.32470/CCN.2019.1219-0.
- [P10] Saitis, C., Fritz, C., Scavone, G., & Giordano, B.L. (2013). Evaluating violin quality: A comparison of player reliability in constrained vs unconstrained tasks. In R. Bresin, & A. Askenfelt (Eds.), *Proc. of the Stockholm Music Acoustics Conference 2013 (SMAC 2013)*, 109–114. Stockholm, Sweden.
- [P11] Saitis, C., Fritz, C., Guastavino, C., Giordano, B.L., & Scavone, G. (2013). Perceptual evaluation of violins: A comparison of intra-individual agreement in playing vs. listening tasks for the case of richness. In *Proc. of the 21st International Congress on Acoustics (ICA-2013)*. Montreal, Quebec, Canada. doi:10.1121/1.4800060.
- [P12] Saitis, C., Fritz, C., Giordano, B.L., & Scavone, G. (2012). Bridge admittance measurements of 10 preference-rated violins. In *Proceedings of Acoustics 2012*. Nantes, France.
- [P13] Saitis, C., Giordano, B.L., Fritz, C., & Scavone, G. (2011). Aspects of experimental design for the perceptual evaluation of violin qualities. *Canadian Acoustics*, 39, 134–135.
- [P14] Saitis, C., Giordano, B.L., Fritz, C., & Scavone, G. (2011). Investigating the origin of interindividual differences in the preference for violins. In *Proceedings of the Forum Acusticum 2011*, 497–501. Aalborg, Denmark.
- [P15] Giordano, B.L., Avanzini, F., Wanderley, M., & McAdams, S. (2010). Multisensory integration in percussion performance. In *Actes du 10eme Congrès Français d'Acoustique, Lyon*, [CD-ROM]. Paris, France: Société Française d'Acoustique.
- [P16] McAdams, S., & Giordano, B.L. (2010). La psychoacoustique dévoile le potentiel musical du timbre. In *Actes du 10eme Congrès Français d'Acoustique, Lyon*, [CD-ROM]. Paris, France: Société Française d'Acoustique.
- [P17] Fischer, M.H., Riello, M., Giordano, B.L., & Rusconi, E. (2010). Singing numbers... in cognitive space. In A. Pease, M. Guhe, & A. Smaill (Eds.), *Proc. of the International Symposium on Mathematical Practice and Cognition, AISB Convention*, 16–17. De Montfort University, Leicester, UK.
- [P18] Bouchara, T., Giordano, B.L., Frissen, I., Katz, F.G., & Guastavino, C. (2010). Effect of signal-to-noise ratio and visual context on environmental sound identification. In *Audio Engineering Society Convention 128*.
- [P19] Visell, Y., Cooperstock, J., Giordano, B.L., Franinovic, K., A. Law, A., McAdams, S., Jathal, K., & Fontana, F. (2008). A vibrotactile device for display of virtual ground materials in walking. In M. Ferre (Ed.), *Lecture Notes in Computer Science – Haptics: Perception, Devices and Scenarios – Proc. of EuroHaptics 2008*, volume 5024, 420–426. Berlin, Germany: Springer-Verlag.
- [P20] Peters, N., Giordano, B.L., Kim, S., Braasch, J., & McAdams, S. (2008). Predicting perceived off-center sound degradation in surround loudspeaker setups for various multichannel microphone techniques. In *Audio Engineering Society Convention 125*.
- [P21] Giordano, B.L., McAdams, S., & McDonnell, J. (2007). Acoustical and conceptual information for the perception of animate and inanimate sound sources. In G.P. Scavone (Ed.), *Proc. of the 13th International Conference on Auditory Display (ICAD2007)*, 173–180. Montréal, Canada: Schulich School of Music, McGill University.
- [P22] Giordano, B.L., & Petrini, K. (2003). Hardness recognition in synthetic sounds. In R. Bresin (Ed.), *Proc. of the Stockholm Music Acoustics Conference (SMAC03)*, volume 2, 611–614. Stockholm, Sweden.

- [P23] Sinico, M., & Giordano, B.L. (2003). Sul ruolo funzionale di un suono tra due suoni. In *Figure e sfondo – Temi e variazioni per Paolo Bozzi*, 311–314. Padova, Italy: CLEUP.
- [P24] Giordano, B.L., & Rocchesso, D. (2003). The effect of inharmonicity on the perceived quality of piano tones. In *Proc. of the XIII Colloquium on Musical Informatics (CIM)*, 177–179. L'Aquila, Italy.

Abstracts in conference proceedings

- [A1] Esposito, M., Ferreyra, C., Weidler, T., Plegat, M., Marinato, G., Formisano, E., & Giordano, B.L. (2026). Linking human auditory representations and deep neural networks through multiscale temporal processing. In *8th Salzburg Mind–Brain Annual Meeting (SAMBA 2026)*. Salzburg, Austria. Poster 17.
- [A2] Marinato, G., Plegat, M., Ferreyra, C., de Araújo Vitória, M., Esposito, M., Formisano, E., & Giordano, B.L. (2026). Sound identifiability modulates acoustic-to-semantic transformations in MEG responses to natural sounds. In *8th Salzburg Mind–Brain Annual Meeting (SAMBA 2026)*. Salzburg, Austria. Poster 49.
- [A3] Giordano, B.L. (2026). Characterizing natural sound representations with magnetoencephalography. In *8th Salzburg Mind–Brain Annual Meeting (SAMBA 2026)*. Salzburg, Austria. Invited talk.
- [A4] Plegat, M., Marinato, G., Ferreyra, C., de Araújo Vitória, M., Esposito, M., Schön, D., Formisano, E., & Giordano, B.L. (2026). A temporal hierarchy of sound-driven semantic representations in MEG responses to natural sounds. In *Organization for Human Brain Mapping (OHBM) 2026 Annual Meeting*. Bordeaux, France. Poster 1298.
- [A5] Esposito, M., Ferreyra, C., Weidler, T., Plegat, M., Marinato, G., Formisano, E., & Giordano, B.L. (2026). Aligning multiscale time-resolved sound-classification networks with human magnetoencephalography. In *Organization for Human Brain Mapping (OHBM) 2026 Annual Meeting*. Bordeaux, France. Poster 0759.
- [A6] Giordano, B.L. (2025). The perceptual and cerebral representation of natural sounds. In *8th International Conference on Auditory Cortex*. Maastricht, the Netherlands.
- [A7] Araújo Vitória, M., Plegat, M., Marinato, G., Ferreyra, C., Esposito, M., Herff, C., Giordano, B.L., & Formisano, E. (2025). Acoustic-to-semantic transformation of natural sounds: The role of frontal areas. In *8th International Conference on Auditory Cortex*. Maastricht, the Netherlands.
- [A8] Esposito, M., Valente, G., Giordano, B.L., & Formisano, E. (2025). Modeling auditory cortical processing: A deep learning approach to hierarchical auditory integration. In *8th International Conference on Auditory Cortex*. Maastricht, the Netherlands.
- [A9] Lopez-Madrona, V., Giroud, J., Mercier, M., Giordano, B.L., Trebuchon, A., Poeppel, D., & Morillon, B. (2025). Theta and gamma rhythms in the human auditory cortex are evoked by speech. In *8th International Conference on Auditory Cortex*. Maastricht, the Netherlands.
- [A10] Plegat, M., Marinato, G., Ferreyra, C., Araújo Vitória, M., Esposito, M., Schön, D., Formisano, E., & Giordano, B.L. (2025). Temporal dynamics of natural sounds representation in the human brain. In *8th International Conference on Auditory Cortex*. Maastricht, the Netherlands.
- [A11] Ferreyra, C., Plegat, M., Marinato, G., Araújo Vitória, M., Esposito, M., Formisano, E., Artières, T., & Giordano, B.L. (2025). Auditory redundant and synergistic representations between brain and artificial neural networks revealed by information-theoretic decomposition. In *8th International Conference on Auditory Cortex*. Maastricht, the Netherlands.
- [A12] Marinato, G., Ferreyra, C., Plegat, M., Esposito, M., Formisano, E., & Giordano, B.L. (2025). Learned semantic associations from real-world events co-occurrences influence the auditory perceptions of naturalistic sounds scenes. In *8th International Conference on Auditory Cortex*. Maastricht, the Netherlands.
- [A13] Giordano, B.L., Whiting, C., Kriegeskorte, N., Kotz, S.A., Gross, J., & Belin, P. (2019). The representational dynamics of perceived voice emotions evolve from categories to dimensions. In *25th Annual Meeting of the Organization for Human Brain Mapping*. Rome, Italy.
- [A14] Cao, Y., Summerfield, C., Park, H., Giordano, B.L., & Kayser, C. (2019). Causal inference in the multisensory brain. In *25th Annual Meeting of the Organization for Human Brain Mapping*. Rome, Italy.
- [A15] Brovelli, A., Basanisi, R., Giordano, B.L., & Panzeri, S. (2018). Brain network dynamics of arbitrary visuomotor learning. In *Society for Neuroscience Abstracts*.
- [A16] Giordano, B.L., Whiting, C., Kriegeskorte, N., Kotz, S.A., Belin, P., & Gross, J. (2017). Multivariate fmri and meg responses to vocalizations reveal temporally and spatially segregated representations of emotion categories and dimensions. In *6th International Conference on Auditory Cortex*. Banff, Canada.
- [A17] Whiting, C., Giordano, B.L., Belin, P., & Gross, J. (2016). Neural encoding of vocal emotion in multivariate spatiotemporal meg patterns. In *BIOMAG2016*. Seoul, Korea.

- [A18] Lillywhite, A., Glowinski, D., Giordano, B.L., Camurri, A., Crossn, I., Hawkins, S., & Pollick, F. (2015). An fMRI study investigating the contribution of visual features to an intersubject correlation (ISC) of brain activity during observation of a string quartet. *Perception*, 44, 347–348.
- [A19] Fleming, D., Giordano, B.L., Caldara, R., & Belin, P. (2014). Sensitivity to language categories in superior temporal cortices: A multivariate study. In *20th Annual Meeting of the Organization for Human Brain Mapping*. Hamburg, Germany.
- [A20] Salvia, E., Kotz, S.A., Bestelmeyer, P., Pernet, C., Rousselet, G.A., Giordano, B.L., Gross, J., & Belin, P. (2014). Cerebral processing of affective nonverbal vocalizations using MEG and voice morphing: Single-subject GLM analysis. In *5th International Conference on Auditory Cortex*. Magdeburg, Germany.
- [A21] Zang, J., Smith, F.W., Giordano, B.L., Grosbras, M., Rousselet, G.A., & Belin, P. (2014). Lifelong changes in the auditory cortex for voice perception. In *5th International Conference on Auditory Cortex*. Magdeburg, Germany.
- [A22] Fleming, D., Giordano, B.L., Caldara, R., & Belin, P. (2014). Sensitivity to language categories in the auditory cortex of Mandarin- and English- speaking participants. In *5th International Conference on Auditory Cortex*. Magdeburg, Germany.
- [A23] Giordano, B.L., Fleming, D., & Belin, P. (2014). The representational geometry of natural-sound categories in the auditory cortex. In *5th International Conference on Auditory Cortex*. Magdeburg, Germany.
- [A24] Giordano, B.L., Panzeri, S., Ince, R., de Cheveigné, A., Gross, J., & Belin, P. (2013). Phase of low-frequency cortical oscillations encodes sound edges and sound-specific structure. In *19th Annual Meeting of the Organization for Human Brain Mapping*. Seattle, WA.
- [A25] Fleming, D., Giordano, B.L., Caldara, R., & Belin, P. (2013). Mapping the neural correlates of the “language familiarity effect”: A similarity analysis approach. In *19th Annual Meeting of the Organization for Human Brain Mapping*. Seattle, WA.
- [A26] Saitis, C., Fritz, C., Guastavino, C., Giordano, B.L., & Scavone, G. (2012). Investigating consistency in verbal descriptions of violin preference by experienced players. In E. Cambouropoulos, C. Tsougras, P. Mavromatis, & K. Pastiadis (Eds.), *12th International Conference on Music Perception and Cognition (ICMPC2012)*, 875–876. Thessaloniki, Greece.
- [A27] Giordano, B.L., Shafiro, V., & Kharkhurin, A. (2011). Accurate estimation of the temporal dynamics of bouncing events. *Journal of the Acoustical Society of America*, 129, 2594. doi:10.1121/1.3588596.
- [A28] Giordano, B.L., McAdams, S., Kriegeskorte, N., Zatorre, R., & Belin, P. (2010). Categorical neural processing of environmental sounds: activation- and information-based mapping. In *16th Annual Meeting of the Organization for Human Brain Mapping*. Barcelona, Spain.
- [A29] Saitis, C., Scavone, G.P., Fritz, C., & Giordano, B.L. (2010). Evaluating violin quality: How consistent are skilled players? *Journal of the Acoustical Society of America*, 128, 2284. doi:10.1121/1.3508002.
- [A30] Giordano, B.L., McAdams, S., Kriegeskorte, N., Zatorre, R., & Belin, P. (2010). Category specificities in the neural processing of environmental sounds: Activation-based mapping. In *Proc. of the Cognitive Neuroscience Society (CNS) 2010 Annual Meeting*, 128. Montréal, Canada.
- [A31] Belin, P., Giordano, B.L., McAdams, S., Kriegeskorte, N., & Zatorre, R. (2010). Category specificities in the neural processing of environmental sounds: Information-based mapping. In *Proc. of the Cognitive Neuroscience Society (CNS) 2010 Annual Meeting*, 128. Montréal, Canada.
- [A32] Giordano, B.L., Guastavino, C., Murphy, E., Ogg, M., Smith, B.K., & McAdams, S. (2009). Common and distinctive features of similarity: Effect of behavioral method. In *Proc. of the 50th annual meeting of the Psychonomic Society*, Abstract number 2007. Boston, MA.
- [A33] Giordano, B.L., Avanzini, F., Wanderley, M., & McAdams, S. (2009). Integrating nonspatial, nontemporal multisensory information in action-based perception. In *Proc. of the 10th International Multisensory Research Forum (IMRF09)*, 64–65. New York, NY.
- [A34] Gingras, B., Lagrandeur-Ponce, T., Giordano, B.L., & McAdams, S. (2008). Effect of expressive intent, performer expertise, and listener expertise on the perception of artistic individuality in organ performance. In *Proc. of the 10th International Conference on Music Perception and Cognition (ICMPC2008)*. Sapporo, Japan.
- [A35] Giordano, B.L., McAdams, S., Visell, Y., Cooperstock, J., Yao, H.Y., & Hayward, V. (2008). Non-visual identification of walking grounds. *Journal of the Acoustical Society of America*, 123, 3412. doi:10.1121/1.2934136.

- [A36] Giordano, B.L., McAdams, S., Crosato, P., Avanzini, F., Casciato, C., Sinclair, S., & Wanderley, M.M. (2008). Action-based multisensory integration in striking events. *Journal of the Acoustical Society of America*, 123, 3178. doi:10.1121/1.2933270.
- [A37] Lagrange, M., Giordano, B.L., Depalle, P., & McAdams, S. (2008). Objective quality measurement of the excitation of impact sounds in a source/filter model. *Journal of the Acoustical Society of America*, 123, 3746. doi:10.1121/1.2935296.
- [A38] Castiello, U., Tubaldi, F., Ansuini, C., Giordano, B.L., & Grassi, M. (2007). Multisensory integration during grasping movements. In *Proc. XV Congresso Società Italiana di Psicofisiologia*. Pisa, Italy.
- [A39] Giordano, B.L., McDonnell, J., & McAdams, S. (2007). Cognitive biases in estimating the similarity of environmental sounds. *Abstracts of the Psychonomic Society*, 12.
- [A40] Giordano, B.L., McAdams, S., & Rocchesso, D. (2007). The object of perception in impacted sound sources: The role of perceptual accessibility and information accuracy. *Journal of the Acoustical Society of America*, 121, 3166. doi:10.1121/1.4782261.
- [A41] McAdams, S., Giordano, B.L., Susini, P., Peeters, G., & Rioux, V. (2006). A meta-analysis of acoustic correlates of timbre dimensions. *Journal of the Acoustical Society of America*, 120, 3275–3276. doi:10.1121/1.4777215.
- [A42] Giordano, B.L., & McAdams, S. (2006). The relevance of source properties to timbre perception: evidence from previous studies on musical instrument tones. In M. Baroni, A.R. Addessi, R. Caterina, & M. Costa (Eds.), *Proc. of the 9th International Conference on Music Perception and Cognition (ICMPC2006)*. Bologna, Italy.
- [A43] Giordano, B.L., & Bresin, R. (2006). Walking and playing: what's the origin of emotional expressiveness in music? In M. Baroni, A.R. Addessi, R. Caterina, & M. Costa (Eds.), *Proc. of the 9th International Conference on Music Perception and Cognition (ICMPC2006)*. Bologna, Italy.
- [A44] Rusconi, E., Giordano, B.L., Casey, A., Umiltà, C., & Butterworth, B. (2006). Horizontal and vertical dimensions in the spatial representation of pitch height. In M. Baroni, A.R. Addessi, R. Caterina, & M. Costa (Eds.), *Proceedings of the 9th International Conference on Music Perception and Cognition (ICMPC 2006)*. Bologna, Italy.
- [A45] McAdams, S., & Giordano, B.L. (2006). Generalizing timbre space data across stimulus contexts: The meta-analytic approach. *Journal of the Acoustical Society of America*, 119, 3395. doi:10.1121/1.4786681.
- [A46] Giordano, B.L., & Bressan, P. (2003). Lights, shadows, and filters: a parametric study of neon brightness spreading. *Perception (ECVP Abstract Supplement)*, 32, 152.

Scientific service and collective responsibilities

Blurb Associate Editor (3 journals), institutional governance and committees (including platform and doctoral-related committees), recruitment and promotion committees, PhD and HDR juries, conference chairing and program committees, evaluation for major funding agencies, and journal peer review across more than 50 journals since 2006.

Editorial responsibilities

- Associate editor: *Frontiers in Neuroscience and Psychology: Auditory Cognitive Neuroscience*.
- Associate editor: *Auditory Perception & Cognition* (Taylor & Francis).
- Associate editor: *Brain Topography* (Springer Nature).

Institutional governance and committees

- 2026 - NeuroSchool committee for allocation of PhD extension funding, Aix Marseille Université, Marseille, France. 12–13 February 2026
- 2025– - Steering group member, Optically-pumped magnetometers (OPM) platform, Institut de Neurosciences des Systèmes, Aix Marseille Université, Marseille, France.
- 2023–2026 - PhD follow-up committee (CSI – comité de suivi individuel de these): Clement Sauvage, Institut de Neurosciences des Systèmes, Aix Marseille Université, Marseille, France.
- 2023 - PhD follow-up committee (CSI – comité de suivi individuel de these): Paul Robert-Lalande, Institut de Neurosciences des Systèmes, Aix Marseille Université, Marseille, France.

- 2015–2016 - Research Planning and Strategy Committee, University of Glasgow.
- 2015–2016 - Representative of Institute of Neuroscience and Psychology in Postdoctoral Researcher Forum, University of Glasgow.

Organization of events

- Invited seminars** - Prof. Laurie Heller, Institut des Neurosciences de La Timone. Marseille, France, 20 October 2025. Role: local organizer and funding.
- Prof. Elia Formisano, Institut des Neurosciences de La Timone. Marseille, France, 5 March 2024. Role: local organizer and funding.

- Cross-team scientific coordination** - Joint NASCE lab retreat: Natural Sounds Cognition and Neuroscience (NaSCog, Institut des Neurosciences de La Timone, CNRS and Aix-Marseille University, leader: Bruno L. Giordano) and Auditory Cognition Research group (Maastricht University, leader: Elia Formisano). Carry-le-Rouet, France, organized for 7–9 October 2026. Role: organizer and co-funding.
- *Representation in brains and machines*, BraiNets thematic days, Institut de Neurosciences de la Timone, Marseille, France, 26 June 2026. Role: organizer.
- *Analysis of distances in neuroscience*, BraiNets thematic days, Institut de Neurosciences de la Timone, Marseille, France, 6 November 2025. Role: organizer and speaker.
- Joint lab meeting: Audition-to-semantics (Aud2Sem, INT, CNRS & AMU, leader: B.L. Giordano); Neural Bases of Communication (BANCO, INT, CNRS & AMU, leader: Pascal Belin); Dynamics of Communication and Auditory Processes (D-CAP, INS, Inserm & AMU, leader: Daniele Schön). Institut des Neurosciences de La Timone, Marseille, France, 20 October 2025. Role: organizer and funding.
- Joint lab retreat: Neural bases of communication (BANCO, Institut des Neurosciences de La Timone, CNRS and Aix-Marseille University, leader: Pascal Belin) and Auditory Cognition Research group (Maastricht University, leader: Elia Formisano). Posada, Italy, 29 April – 2 May 2018. Role: co-organizer and co-funding.

- Workshops** - Workshop on multivariate analysis of fMRI data, Institute of Neuroscience and Psychology, University of Glasgow. Glasgow, UK, 17-18 September 2012. Role: co-organizer.

Hosted research visits

- July–August 2026 - Prof. Sonja A. Kotz (Maastricht University), hosted at the Institut de Neurosciences de la Timone, Marseille, 15 June–31 July 2026. Role: scientific host; visit funded through the CNRS component of ERC Synergy NASCE. Collaboration on AURIS test-battery development and on the clinical and SEEG framework for characterising future patients with acquired auditory-recognition disorders.
- May–June 2024 - Michele Esposito (Maastricht University), research visit to INT within BraiNets. Work with B.L. Giordano on hyperparameter tuning of the multiscale recurrent network for sound recognition, later reported in a 2026 bioRxiv preprint.

Recruitment and promotion committees

- 2024 - Selection committee for lecturer-level position (maître de conférences) at Aix-Marseille University on Acoustics and Digital Audio Signal Processing (2024).
- 2017 - Tenure promotion committee, Department of Psychology, Carnegie Mellon University, Pittsburgh, Pennsylvania, USA.

Thesis examination and HDR juries

- 2026 - Doctoral thesis: Mr. Lidongsheng Xing, “Identifying Sound Features from Brain Activity”, Maastricht University, The Netherlands.
- 2023 - HDR thesis (habilitation à diriger la recherche): Dr. Benedikt Zoefel, “Rhythmic neural and perceptual dynamics in the auditory system”, Université Paul Sabatier – Toulouse III, Toulouse, France.

- 2022 - Doctoral thesis: Dr. James Plumridge, “Language knowledge and visual speech processing: An EEG investigation”, School of Psychology, Faculty of Health, Deakin University, Melbourne, Australia.
- Doctoral thesis: Dr. Victor Rosi, “The metaphors of sound: From semantics to acoustics”, Institut de Recherche et Coordination Acoustique/Musique (STMS-IRCAM-CNRS), Paris.
- 2019 - Doctoral thesis: Ms. Vittoria De Angelis, “Computational neuroimaging of real-life listening”, Maastricht University, The Netherlands.
- 2016 - HDR thesis (habilitation a diriger la recherche): Dr. Guillaume Lemaitre, “Everyday sound: A journey through perception, cognition, interaction, and imitation of sounds that surround us”, Université Pierre et Marie Curie, Paris.
- 2015 - Doctoral thesis: Dr. Eric Boyer, “Continuous auditory feedback for sensorimotor learning”, Institut de Recherche et Coordination Acoustique/Musique (STMS-IRCAM-CNRS), Paris.
- 2014 - Doctoral thesis: Dr. Hamed Nili, “Understanding brain representational geometries”, MRC Cognition and Brain Sciences Unit, University of Cambridge.

Conference leadership

- Chair** - Auditory Perception and Cognition – Music and the Brain session, 14th International Symposium on Computer Music Multidisciplinary Research (CMMR 2019), 2019, Marseille, France.
- Perception session, 10th Sound and Music Computing Conference (SMC 2013), 2013, Stockholm, Sweden.
- Program committee** - 14th International Symposium on Computer Music Multidisciplinary Research (CMMR 2019), 2019, Marseille, France.
- 14th Sound and Music Computing Conference (SMC 2017), 2017, Helsinki, Finland.
- XXI Colloquium on Music Informatics (XXI CIM 2016), 2016, Cagliari, Italy.
- 13th Sound and Music Computing Conference (SMC 2016), 2016, Hamburg, Germany.
- 12th International Symposium on Computer Music Multidisciplinary Research (CMMR 2016), 2016, São Paulo, Brasil.
- 11th International Symposium on Computer Music Multidisciplinary Research (CMMR 2015), 2015, Plymouth, UK.
- XX Colloquium on Music Informatics (XX CIM 2014), 2013, Rome, Italy.
- Stockholm Music Acoustics Conference 2013 (SMAC 2013) jointly with 10th Sound and Music Computing Conference (SMC 2013), 2013, Stockholm, Sweden.
- 10th International Symposium on Computer Music Multidisciplinary Research (CMMR 2013), 2013, Marseille, France.
- 9th International Symposium on Computer Music Modeling and Retrieval (CMMR 2012), 2012, London, UK.
- 9th Sound and Music Computing Conference (SMC 2012), 2012, Copenhagen, Denmark.
- 8th Sound and Music Computing Conference (SMC 2011), 2011, Padova, Italy.
- 7th Sound and Music Computing Conference (SMC 2010), 2010, Barcelona, Spain.
- 7th International Symposium on Computer Music Modeling and Retrieval (CMMR 2010), 2010, Malaga, Spain.

Evaluation for funding agencies

- National Science Foundation (NSF; 2023)
- Biotechnology & Biological Sciences Research Council (BBSRC; 2018)
- Institut de Recherche Biomédicale des Armées (French Armed Forces Biomedical Research Institute; IRBA; 2016)

- Economic & Social Research Council (ESRC; UK; 2015)
- National Institute of Health (NIH): Special Emphasis Panel (SEP) ZRG1 BBBP-T (52) RFA-EY-13-001 Basic Behavioral Research on Multisensory Processing Study Section (2013)

Mock interview committees

- 2026 - Mock interview for ERC Advanced proposal – Prof. Sonja A. Kotz (ERC-2025-SyG)

Journal peer review

Acta Acustica United with Acustica; Acta Psychologica; Applied Acoustics; Applied Intelligence; Applied Sciences; Attention, Perception, & Psychophysics; Behavior Research Methods; Brain and Language; Brain Research; Cerebral Cortex; **Cognition**; Cognitive and Behavioral Neurology; Cognitive Psychology; Communications Biology; Communications Psychology; Cortex; Current Research in Neurobiology; Current Zoology; European Journal of Neuroscience; Experimental Brain Research; Frontiers in Neuroscience – Auditory Cognitive Neuroscience; Frontiers in Psychology – Environmental Psychology; Frontiers in Psychology – Perception Science; Frontiers in Systems Neuroscience; Hearing Research; Human Brain Mapping; IEEE Access; IEEE Transactions on Haptics; International Journal of Audiology; International Journal of Human-Computer Studies; iScience; Journal of Experimental Psychology: General; Journal of Experimental Psychology: Human Perception and Performance; Journal of the Acoustical Society of America; Journal of the Acoustical Society of America – Express Letters; Journal of the Audio Engineering Society; Journal on Multimodal User Interfaces; Music Perception; Music and Science; Multivariate Behavioral Research; **Nature Communications**; NeuroImage; Neuropsychologia; Neuroscience Letters; New Ideas in Psychology; PLOS Biology; PLOS ONE; Perception; **Proceedings of the National Academy of Sciences of the United States of America (PNAS)**; Progress in Neurobiology; Psychological Science; Psychology of Music; Quarterly Journal of Experimental Psychology; SciTechnol – The Journal of Otology & Rhinology; **The Journal of Neuroscience**; Transactions on Applied Perception.

Conference proceedings review

- 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP2023).
- Annual Meeting of the Organization on Human Brain Mapping (HBM; 2011–2015).
- Eurohaptics 2014, Versailles, France.
- Haptic Audio Interaction Design conference (HAID 2010), 2010, Ballerup, Denmark.
- Enactive 2008, 2008, Pisa, Italy.
- Fifth International Symposium on Computer Music Modeling and Retrieval (CMMR 2008), 2008, Copenhagen, Denmark.
- Seventh European Gesture Workshop (GW 2007), 2007, Lisbon, Portugal.
- Ninth International Conference on Digital Audio Effects (DAFx-06), 2006, Montréal, QC, Canada.

Funding

Grants as PI/co-PI

- 2025–2031 European Research Council (ERC), Synergy program: ERC-2024-SyG: NASCE (Proj. 101167313, PI with corresponding PI E. Formisano). € 8,622,811 (contribution to CNRS partner = € 3,544,851).
- 2022–2026 French National Research Agency (ANR-21-CE37-0027). € 543,253.
- 2020–2022 Institute of Language, Communication and the Brain (ILCB) Equipe Projet. € 50,000.
- 2015–2018 Biotechnology and Biological Sciences Research Council (BBSRC). Researcher Co-I (non-permanent PI role) with J. Gross (PI), Institute of Neuroscience and Psychology, University of Glasgow, Glasgow, UK, £ 743,155.

Fellowships and bursaries

- 2012-2014 Marie Curie Intra-European Fellowship for Career Development (IEF), Institute of Neuroscience and Psychology, Centre for Cognitive Neuroimaging, University of Glasgow, Glasgow, UK, € 278,807.
- 2011-2013 Erasmus Mundus Postdoctoral Fellowship (Auditory Cognitive Neuroscience Student Exchange Network), Institute of Neuroscience and Psychology, Centre for Cognitive Neuroimaging, University of Glasgow, Glasgow, UK (with P. Belin), € 43,400.
- 2005 COST-IST action “ConGAS”, Short Term Scientific Mission to Department of Speech, Music and Hearing (TMH), Royal Institute of Technology (KTH), Stockholm, Sweden (with R. Bresin; PI: N. Bernardini), € 1,500.
- 2003–2004 Marie Curie Fellow at Department of Speech, Music and Hearing (TMH), Royal Institute of Technology (KTH), Stockholm, Sweden., financed by Marie Curie Training Site Fellowship, FP5 contract number HPMT-CT-2000-00119 (with S. Ternström [PI]), € 7,200.
- 2001–2004 Doctoral Scholarship, Assessorato della Pubblica Istruzione Beni Culturali, Informazione Spettacolo e Sport, Regione Autonoma della Sardegna, [Council for Public Education, Cultural Heritage, Information, Entertainment and Sports, Autonomous Province of Sardinia], Italy, € 30,984.

Grants as external collaborator

- 2021-2026 Dutch Research Council (NWO), with E. Formisano (PI), Department of Cognitive Neuroscience, Faculty of Psychology and Neuroscience, Maastricht University, Maastricht, Netherlands, € 725,000.

Competitive fellowships hosted under my supervision

- 2026–2028 Marie Skłodowska-Curie Actions Postdoctoral Fellowship (HORIZON-MSCA-2025-PF), *SONATA: Brain-Aligned Neural Architectures for Real-Time Sound Recognition* (Project 101279402; € 226,420.56; 24 months; started 1 May 2026). Fellow: Michele Esposito; beneficiary: Aix-Marseille Université; host supervisor: B.L. Giordano. I contributed to proposal development.

Prizes and awards

- 2023 Travel grant: Acoustical Society of America. € 1,350
- 2018 Allocation d’Installation Chercheurs, Ville de Marseille. Prize awarded annually by the Marseille municipality to researchers moving to Marseille that were previously hired outside the Aix-Marseille University. The prize is awarded based on the curriculum vitae and a scientific project described by the candidate, and is attributed by a jury of local scientists. € 3,000

Co-authored grants

- 2010–2011 Psychomechanics of aerodynamic sounds (with G. Scavone [PI], S. McAdams, L. Mongeau, McGill University), CIRMMT Strategic Innovation Fund, \$10,000.
- 2010–2015 Auditory perception and cognition of sources, events and sequences (with S. McAdams [PI], McGill University), NSERC Discovery Grant, \$250,000.
- 2006–2009 Haptics, sound and interaction in the design of enactive interfaces (with M. M. Wanderley [PI], V. Hayward, S. McAdams, G. Scavone, P. Depalle and C. Guastavino, McGill University), NSERC Special Research Opportunity, \$479,651.

International research projects

- 2014–2016 Audio-Haptic modalities in Musical Interfaces (with S. Papetti [PI], see [project webpage under development](#)).
- 2008–2011 Sonic Interaction Design (SID), Cost IC0601 Action (with D. Rocchesso [PI], see <http://www.cost-sid.org/>).
- 2008–2011 Natural Interactive Walking (NIW), Seventh Framework Programme, Project no 222107 (with J. R. Cooperstock and F. Fontana [PI], see <http://www.niwproject.eu/>).

2001–2002 The Sounding Object, Project number IST-2000 25287 (with D. Rocchesso [PI], and G. B. Vicario, see <http://www.soundobject.org/>).

Supervision

Current team: main supervisor

- Research group** NaSCog (Natural Sounds Cognition and Neuroscience), sub-group of BraiNets team at INT.
- March 2026– Mia-Sara Bertovic (doctoral student, INT/INS, CNRS and AMU, Marseille), with Aurelie Bidet-Caulet. Topic: Affective dimensions of natural sound responses in the human brain and body.
- October 2025– Michele Esposito (**postdoctoral fellow**, INT, CNRS and AMU, Marseille). Topic: brain-aligned neural architectures for real-time sound recognition and artificial-intelligence models of electrophysiological responses to natural sounds. Since 1 May 2026, **MSCA Postdoctoral Fellow for SONATA (Project 101279402) under my supervision**.
- October 2025– Margarida Moreira (doctoral student, INT, CNRS and AMU, Marseille). Topic: Stimulus-driven attention in the cerebral processing of natural sound scenes.
- November 2023– Christian Ferreyra (doctoral student, INT, CNRS and AMU, Marseille), with Thierry Artieres. Topic: Information-theoretic analyses of MEG data and artificial neural networks.
- December 2022– Giorgio Marinato (**postdoctoral fellow**, INT, CNRS and AMU, Marseille). Topic: Electrophysiology of natural sound representation in the human brain.
- October 2022– Marie Plegat (doctoral student, INT/INS, CNRS and AMU, Marseille), with Daniele Schön. Topic: Spatio-temporal dynamics of natural sound representation in magnetoencephalography data. **Thesis defense scheduled for 1 February 2027. Examining committee: Dr. Daniel Pressnitzer (CNRS & Ecole normale supérieure); Dr. Anne Caclin (Lyon Neuroscience Research Center); Prof. Jennifer Bizley (UCL, London, UK)**

Past supervision: main supervisor

- 2025–2026 Sara Carta (**postdoctoral fellow**, INT, CNRS and AMU, Marseille). Topic: Saliency and semantic congruence effects on the cerebral representation of natural sound scenes.
- 2021 Dylan Ginet (Master's student, Aix-Marseille University, Marseille). Topic: computational models of natural sound perception.
- 2021 Lila Barrillot (Master's student, Aix-Marseille University, Marseille). Topic: physiological correlates of perceived voice emotions.
- 2020 Thibaut Piquemal (Master's student, Aix-Marseille University, Marseille). Topic: acoustics and semantics of natural sounds.

Past co-supervision

- 2022 Martin Bremm (Master's student, Department of Cognitive Neuroscience, Faculty of Psychology and Neuroscience, Maastricht University, Maastricht, The Netherlands), with Elia Formisano and Michel Dumontier. Topic: Development of a novel ontology of natural sounds.
- 2021–2024 Michele Esposito (doctoral student, Department of Cognitive Neuroscience, Faculty of Psychology and Neuroscience, Maastricht University, Maastricht, The Netherlands), with Elia Formisano. Topic: Deep neural network models of natural sound processing.
- 2021 Ricardo de Miranda Azevedo (**post-doctoral fellow**, Institute of Data Science, Faculty of Science and Engineering, Maastricht University, Maastricht, The Netherlands), with Elia Formisano and Michel Dumontier. Topic: Review of taxonomies and ontologies of natural sounds.
- 2019–2022 Charly Lamothe (doctoral student, Institut de Neuroscience de la Timone, CNRS and Aix-Marseille University, Marseille), with Pascal Belin and Thierry Artieres. Topic: RSA assessment of the representation of deep learning networks in the human and primate brain.

- 2016–2019 Yinan Cao (doctoral student, Department of Experimental Psychology, Oxford University), co-supervised with Christoph Kayser and Christopher Summerfield. Topic: computational modelling of audiovisual integration in electrophysiology (MEG).
- 2015–2018 Caroline Whiting (**post-doctoral fellow**, Institute of Neuroscience and Psychology, University of Glasgow), co-supervised with Joachim Gross and Pascal Belin. Topic: fMRI and MEG imaging of vocal affect.
- 2016–2017 Alexandra Vossen (research assistant, Institute of Neuroscience and Psychology, University of Glasgow), co-supervised with Joachim Gross. Topic: EEG imaging of vocal affect.
- 2013–2015 David Fleming (doctoral student, Institute of Neuroscience and Psychology, University of Glasgow), co-supervised with Pascal Belin. Topic: cortical processing of voices.
- 2013–2015 Yinan Cao (undergraduate student, Music Technology, McGill University), co-supervised with Stephen McAdams. Topic: audio-haptic integration in percussion performance.
- 2014 Jingting Zhang (**post-doctoral fellow**, Institute of Neuroscience and Psychology, University of Glasgow), co-supervised with Pascal Belin. Topic: effects of aging on the cortical encoding of vocal affect.
- 2013–2014 Emilie Salvia (**post-doctoral fellow**, Institute of Neuroscience and Psychology, University of Glasgow), co-supervised with Pascal Belin. Topic: encoding of vocally-expressed emotions in MEG data.
- 2010–2014 Charalampos Saitis (doctoral student, Music Technology, McGill University), co-supervised with Gary Scavone. Topic: perceptual quality of violin tones.
- 2010 Alexander Porres (visiting doctoral student from University of Sao Paulo, Brazil), co-supervised with Sean Ferguson. Topic: auditory perception of consonance and dissonance.
- 2009–2010 Yon Visell (doctoral student, Electrical and Computer Engineering, McGill University), co-supervised with Jeremy R. Cooperstock. Topic: integration of multisensory information during locomotion.
- 2009 Tifanie Bouchara (visiting doctoral student, Computer Sciences Laboratory for Mechanics and Engineering Sciences - CNRS, Orsay, France), co-supervised with Catherine Guastavino. Topic: effect of visual context on the identification of environmental sounds presented in noise.
- 2008 Mattson Ogg (undergraduate student, Psychology, McGill University), co-supervised with Stephen McAdams and Catherine Guastavino. Topic: methodological study of behavioral techniques for estimation of dissimilarity.
- 2007 John McDonnell (undergraduate student, Cognitive Sciences, McGill University), co-supervised with Stephen McAdams. Topic: cognitive processing of environmental sounds.

Long-term mentoring and career development

- 2013–2026 Yinan Cao: mentoring from undergraduate research and doctoral co-supervision through later career development, including preparation of ERC and CNRS CRCN applications; currently listed in the Professor category at the Institute of Psychology, Otto von Guericke University Magdeburg, Germany.

Teaching

- 2021 **Representational similarity analysis of neuroimaging and electrophysiology data**, Summer School of the Institute of Language, Communication and the Brain (ILCB), Marseille, France.
- 2014 **fMRI research methods**, Research Methods II (MSc), Institute of Neuroscience and Psychology, University of Glasgow.
- 2014 **Auditory cognitive neuroscience**, Visual perception and cognition (MSc), Institute of Neuroscience and Psychology, University of Glasgow.
- 2013 **Evaluation methods in sonic interaction; perception and brain imaging of sounds**, Sound and Music Computing Summer School, KTH Royal Institute of Technology, Stockholm, Sweden.

- 2013 **Behavioral methods for the study of perception and sensation**, Research Methods II (MSc), Institute of Neuroscience and Psychology, University of Glasgow.
- 2013 **Voice and speech perception**, Visual perception and cognition (MSc), Institute of Neuroscience and Psychology, University of Glasgow.
- 2013 **Auditory perception**, Visual perception and cognition (MSc), Institute of Neuroscience and Psychology, University of Glasgow.
- 2012 **fMRI design and analysis**, Brain imaging methods (MSc), Institute of Neuroscience and Psychology, University of Glasgow.
- 2010 **Design and analysis of behavioral experiments**, Workshop, CIRMMT, McGill University.
- 2010 **Fundamentals of Matlab programming**, Workshop, CIRMMT, McGill University.
- 2006 **Sound source perception**, Artificial Intelligence, Electrical and Computer Engineering, McGill University.
- 2003 **Adaptive psychophysical procedures**, Doctoral course lecture, Department of Psychology, University of Padova, Italy.
- 2002 **Auditory perception**, Psychology of Perception, Faculty of Multimedia Science and Technology, University of Udine, Italy.
- 2002 **Scaling: elements of psychophysics with applications to human-computer interaction**, Doctoral course lecture, Department of Informatics, University of Verona, Italy.
- 1999–2003 **Auditory perception**, Psychology of Perception, Faculty of Psychology, University of Padova, Italy.

Invited talks & oral conference presentations

- 2026 - *Characterizing natural sound representations with magnetoencephalography*, Salzburg Mind Brain Annual Meeting (SAMBA 2026), Salzburg, Austria.
- *The cerebral and perceptual representation of natural sounds*, Institute talk organized by Dr. Ruggero Bassanisi, IMT School for Advanced Studies, Lucca.
- *The cerebral and perceptual representation of natural sounds*, Auditory/Musical Imagery and Perception Conference (AMI&P), Rovereto, Italy.
- *The perceptual and cerebral representation of natural sounds*, Minisymposium on “Sound and Music in the Brain” organized by Prof. Yukiyasu Kamitani (Kyoto University), Kyoto, Japan.
- 2025 - *Assessing model representation, variance partitions and noise ceiling with cross-validated, whitened, and bias-adjusted RSA*, Workshop on Music, Speech, and the Brain, Institut de Neurosciences des Systèmes, Aix Marseille Université, Marseille, France.
- *The perceptual and cerebral representation of natural sounds*, 8th International Conference on Auditory Cortex (ICAC 2025), Maastricht, The Netherlands.
- *The perceptual and cerebral representation of natural sounds*, Neural Oscillations Group (PI: Ole Jensen), Department of Experimental Psychology, Oxford University, UK.
- 2024 - *Auditory-to-semantic computation of natural sounds in the human brain*, workshop on Neuroscience and Music Ontology (NeMO-CAMP 2024), Institut de Neurosciences des Systèmes, Carry-le-Rouet, France.
- *The perceptual and cerebral representation of natural sounds*, 15th Anniversary of CCNi (Centre for Cognitive Neuroimaging), University of Glasgow, Glasgow, UK.
- 2023 - *Intermediate acoustic-to-semantic representations link behavioural and neural responses to natural sounds*, CERIMED (Centre Européen de Recherche en Imagerie Médicale), Faculty of Medicine, Aix Marseille Université, Marseille, France.
- *The perceptual and cerebral representation of natural sounds*, Brain and Cognition Research Center (CerCo), CNRS and Toulouse III University, Toulouse, France.

- *Comparing acoustic, semantic, and deep neural network models of natural sound representation in perceived dissimilarity and cerebral fMRI responses*, 184th Meeting of the Acoustical Society of America, Chicago, IL.
- 2022 - *The perceptual and cerebral representation of natural sounds*, Institut de Neurosciences des Systèmes, Aix Marseille Université, Marseille, France.
- 2020 - *The perceptual and cerebral representation of natural sounds*, Keynote for the XXIII Colloquium on Musical Informatics, Torino, Italy, June 2020 (cancelled due to COVID).
- *The perceptual and cerebral representation of natural sounds*, Laboratoire de Mécanique et d'Acoustique (UMR 7031), CNRS and Aix Marseille University, Marseille, France. February 2020.
- 2019 - *Representational similarity analysis of neuroimaging and electrophysiology data: Strengths, outstanding issues and potential*, 12th Scientific day of the center IRM-INT, Marseille, France.
- *Spatio-temporal dynamics of the cerebral representations of emotion in voice*, OHBM 2019, Rome, Italy.
- *Modelling the perceptual and cerebral representation of natural sounds*, Centre for Human Brain Health, University of Birmingham, UK.
- *Modelling time-varying features in the brain: local representations and transfer*, INT Workshop on Computational Neurophysiology, Institute of Neurosciences de la Timone, Marseille, France.
- 2018 - *Perceiving everyday sound sources: psychoacoustics and neuroimaging*, Cognitive Neuroscience Lab (Prof. C. Kayser), Bielefeld University, Germany.
- 2017 - *Information-theoretic and RSA measurement of cerebral stimulus encoding*, Department of Experimental Psychology, University of Oxford, Oxford, UK.
- *Cerebral encoding of vocal emotions in fMRI and MEG multivariate response patterns*, MEG UK 2017, Oxford, UK.
- *Contributions of local speech encoding and functional connectivity to audio-visual speech integration*, Institut de Neurosciences des Systèmes, Aix Marseille Université, Marseille, France.
- 2016 - *Cerebral encoding of behaviourally estimated vocal emotions*, Institut de Neurosciences de La Timone, Aix Marseille Université, Marseille, France.
- *Perceiving everyday sound sources: psychoacoustics and neuroimaging*, Psychology Department, The Hebrew University of Jerusalem.
- *The dominance of haptics over audition in controlling wrist velocity during striking movements*, Haptics and musical practice workshop, organized by Institute for Computer Music and Sound Technology, Zurich University of the Arts, Zurich, Switzerland.
- *The perceptual and cerebral processing of natural sounds*, NIHR Nottingham Hearing Biomedical Research Unit, Nottingham, UK.
- 2015 - *The perceptual and cerebral processing of natural sounds*, Institut de Recherche et Coordination Acoustique/Musique (STMS-IRCAM-CNRS), Paris, France.
- *The perceptual and cerebral processing of natural sounds*, Music Informatics, Cognition and Acoustic Seminars (MusICA), University of Edinburgh, Edinburgh, UK.
- 2014 - *The perceptual and cerebral processing of natural sounds*, Institut de Neurosciences de La Timone, Aix Marseille Université, Marseille, France.
- *The perceptual and cerebral processing of natural sounds*, Department of Psychology and Neuroscience, Maastricht University, Maastricht, The Netherlands.
- *The perceptual and cerebral processing of natural sounds*, Department of Psychology, University of York, UK.
- *Characterizing the cortical representation of natural sounds using distance matrices*, Institut de Neurosciences de La Timone, Aix Marseille Université, Marseille, France.
- *The perceptual and cerebral processing of natural sounds*, Institute for Computer Music and Sound Technology, Zurich University of the Arts, Zurich, Switzerland.

- 2013 - *Studies on the perceptual and cerebral processing of non-speech sound sources*, Laboratory of Integrative Neuroscience and Cognition (Prof. J. Rauschecker), Georgetown Institute for Cognitive and Computational Sciences, Washington, DC.
- 2012 - *Perceptual and cerebral processing of natural sounds*, School of Social and Health Sciences, University of Abertay Dundee, Dundee, UK.
- *Abstract encoding of auditory objects in cortical activity patterns*, Medical Research Council Cognition and Brain Sciences Unit (MRC-CBU), Representational Similarity Analysis interest group, Cambridge, UK.
- 2011 - *Representational similarity analysis*, Annual Workshop of NSERC-CREATE network in Auditory Cognitive Neuroscience, Montréal, Canada.
- 2010 - *Measuring the perception of unimodal and multisensory naturalistic events*, Medialogy, Aalborg University Copenhagen, Ballerup, Denmark.
- *Cortical processing of environmental sounds*, CIRMMT Student Symposium, Montréal, Canada.
- *Neural correlates of nonlinguistic auditory processing*, Auditory Cognitive Neuroscience Symposium - 2010 International Laboratory for Brain, Music and Sound Research (BRAMS) Scientific Day, Montréal, Canada.
- *Category specificities in the neural processing of environmental sounds: activation- and information-based mapping*, Auditory Cognitive Neuroscience Laboratory, Department of Neuropsychology, McGill University, Montréal, Canada.
- 2009 - *Modeling the perception of musical and natural sounds: acoustics, mechanics and semantics*, CIRMMT Workshop on Timbre, McGill University, Montréal, Canada.
- 2008 - *Audio-haptic integration in action-based perception of impacted objects*, CIRMMT Workshop on Multimodal Integration, McGill University, Montréal, Canada.
- *Perception of the sound source in purely auditory and multisensory contexts*, Music Technology Group, University Pompeu Fabra, Barcelona, Spain.
- *Non-visual identification of walking grounds*, Acoustics08, Paris, France.
- *Perception of sound sources in purely auditory and multisensory contexts*, Department of Psychology, McGill University, Montréal, Canada.
- *Multisensory identification of walking grounds*, CIRMMT Workshop on Multimodal Influences on Perceived Self Motion, McGill University, Montréal, Canada.
- *Perception and categorization of auditory sources and events*, coauthored with Stephen McAdams, Auditory Objects Workshop, University of Toronto, Mississauga, Canada.
- 2007 - *Acoustical and conceptual information for the perception of animate and inanimate sound sources*, International Conference on Auditory Display (ICAD 2007), Montréal, Canada.
- *The object of perception in impacted sound sources: the role of perceptual accessibility and information accuracy*, 151st Meeting of the Acoustical Society of America, Salt Lake City, UT.
- 2006 - *Sound source perception*, Haptics Laboratory, Department of Electrical and Computer Engineering, Faculty of Engineering, McGill University, Montréal, Canada.
- *Sound source perception*, Auditory Cognitive Neuroscience Laboratory, Department of Neuropsychology, McGill University, Montréal, Canada.
- 2005 - *Sound source perception in walking sounds: walkers' emotions, gender, weight and shoes properties*, Department of Speech, Music and Hearing (TMH), Royal Institute of Technology (KTH), Stockholm, Sweden.
- 2002 - *Material recovery from impact sounds: a study in Ecological Acoustics*, Hearing Research Center, Boston University, Boston, MA.
- *Recovering the material of the objects from impact sounds*, Music Perception and Cognition Laboratory, STMS-IRCAM-CNRS, Paris, France.
- *Recovering the material of the objects from impact sounds*, Disappearing Computer Research Atelier Sonification of Hybrid Objects (SOHO), Centro Culturale Candiani, Mestre, Italy.

Dissemination and public engagement

Selected coverage (press releases, institutional/scientific news)

Nature
Neuro-
science
2023 – 2023-03-16 – CNRS (central press office): Press release – L'intelligence artificielle peut-elle égaler le traitement du son par le cerveau ? – <https://www.cnrs.fr/en/press/can-artificial-intelligence-match-how-brain-processes-sound>

– 2023-04-06 – MedicalXpress (Science X): News article / interview – Using deep neural networks to predict how natural sounds are processed by the brain – <https://medicalxpress.com/news/2023-04-deep-neural-networks-natural-brain.html>

– 2023-03-17 – La Gazette du Laboratoire: Brief / press pickup – L'intelligence artificielle peut-elle égaler le traitement du son par le cerveau ? – <https://www.gazettelabo.fr/breves/13187intelligence-artificielle-son-cerveau.html>

– 2023-03-30 – Maastricht University (FPN): Institutional news/blog – Chirp = Bird | Elia Formisano – <https://www.maastrichtuniversity.nl/news/chirp-bird-elia-formisano>

– 2023-04-05 – ILCB (Aix-Marseille): Lab news/blog – Intermediate acoustic-to-semantic representations link behavioral and neural responses to natural sounds – <https://www.ilcb.fr/intermediate-acoustic-to-semantic-representations-link-behavioral-and-neural-responses-to-natural-sounds/>

ERC
Synergy
NASCE – 2024-11-05 – Maastricht University (FPN): Institutional news/blog – Understanding everyday hearing | Elia Formisano | ERC Synergy Grant – <https://www.maastrichtuniversity.nl/news/understanding-everyday-hearing-elia-formisano-erc-synergy-grant>

– 2024-11-05 – CNRS: News / winners page – The winners of 2024 ERC Synergy Grants from the CNRS – <https://www.cnrs.fr/en/update/winners-2024-erc-synergy-grants-cnrs>

– 2024-11 – CNRS-INSB: Laureates page – NASCE (Giordano & Formisano) – <https://www.insb.cnrs.fr/fr/personne/nasce>

– 2024-11-12 – Mission Europe Recherche (Aix-Marseille): Institutional news – New project awarded – ERC Synergy Grant 2024 (NASCE) – <https://mission-europe-recherche.fr/en/news/new-project-awarded-erc-synergy-grant-2024-nasce>

– 2024-11-27 – Neuro-Marseille: Blog/news roundup – All the neuroscience news in December 2024 (mentions NASCE award) – <https://neuro-marseille.org/en/blog/all-the-neuroscience-news-in-december-2024/>

Neuron
2019 – 2019-05-03 – CNRS-INSB (CNRSinfo): Press release (prepared with INT communication staff) – Comment le cerveau combine les stimuli sensoriels – <https://www.insb.cnrs.fr/fr/cnrsinfo/comment-le-cerveau-combine-les-stimuli-sensoriels>

Nature
Human
Behaviour
2021 – 2021-03-15 – CNRS-INSB (CNRSinfo): Press release (prepared with INT communication staff) – Comment notre cerveau traite-t-il les émotions vocales : catégories, ou bien dimensions ? – <https://www.insb.cnrs.fr/fr/cnrsinfo/comment-notre-cerveau-traite-t-il-les-emotions-vocales-categories-ou-bien-dimensions>

Selected press coverage

Nature
Human
Behaviour
2021 – 2021 – La Marseillaise (local newspaper): News feature / interview – On sait enfin comment notre cerveau perçoit les émotions vocales – <https://www.lamarseillaise.fr/societe/on-sait-enfin-comment-notre-cerveau-percoit-les-emotions-vocales-EY6890152>

– 2021 – La Marseillaise (local newspaper): News feature / interview – Nous ignorons encore comment l'émotion perçue est ressentie – <https://www.lamarseillaise.fr/societe/nous-ignorons-encore-comment-l-emotion-percue-est-ressentie-BY6890175>

Selected public engagement

- 2026 - **Festival Explore**, 3rd edition, Marseille, France, 29–31 May 2026. Public engagement around ANR SoundBrainSem (*Représentation des sons naturels dans le cerveau humain : transformation de l'acoustique en sémantique des sources sonores dans l'environnement*), in collaboration with Aix-Marseille Université's Master's students in science communication and the Cellule de Culture Scientifique et Technique.
- *Intermediate acoustic-to-semantic representations link behavioural and neural responses to natural sounds*, QuantumPhotonics Clubhouse Fireside chat, online event on audio-only social media.

Revision: August 31, 2026